

Plot

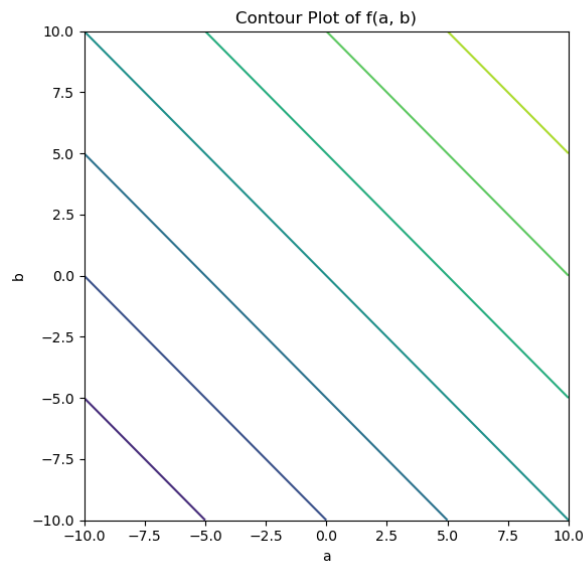


Figure 1: Plot of $f(a, b)$

Code

```
1
2 #
3 # File: __init__.py
4 #
5
6 from project0_py.helpers import *
7
8 ## top-level submission file
9
10
11 ### imports here - TODO: remember to add them to your environment
12 import numpy as np
13 import matplotlib.pyplot as plt
14 ###
15
16
17 def f(a, b):
18     """
19     Function for adding two numbers
20     Args:
21         a(float): first number
22         b(float): second number
23     Returns:
24         c(float): result
```

```
25 """',',
26
27     ## TODO: Fill your code in here
28     c = a + b
29
30     ##
31
32     return c
33
34 # Create a contour plot: see Tutorial 3 or https://matplotlib.org/stable/gallery/images\_contours\_and\_fields/contour\_demo.html
35
36 bottom_limit_a = -10
37 top_limit_a = 10
38 num_steps_a = 100
39
40 bottom_limit_b = -10
41 top_limit_b = 10
42 num_steps_b = 100
43
44 a_vals = np.linspace(bottom_limit_a, top_limit_a, num_steps_a)
45 b_vals = np.linspace(bottom_limit_b, top_limit_b, num_steps_b)
46
47 # Generates a 2D grid of function values
48 A, B = np.meshgrid(a_vals, b_vals)
49 Z = f(A,B)
50
51 # TODO: Replace with the appropriate arguments
52 plt.contour(A,B,Z)
53 plt.title("Contour Plot of f(a,b)")
54 plt.xlabel("a")
55 plt.ylabel("b")
56
57 # show plot
58 plt.show()
```